

DATA.md — what is in this repo, what is not, and how to get the rest

This project uses five datasets. **None of the raw third-party data is redistributed here** — it is external data under its own licence, and this repo pulls it straight from the primary sources with `scripts/fetch_*.py`. What is committed is (a) our own derived artifacts, small enough to live in git, and (b) a precomputed results store shipped as a GitHub Release asset.

The result: you can clone the repo and **run the tests and the offline dashboard immediately**, without downloading a single byte of raw NWP data.

1. What is committed directly in the repo (~42 MB)

These are **our derived products**, not third-party data. Safe to redistribute, small enough for git, and they make the repo runnable.

Path	Size	What it is
<code>data/processed/dataset.parquet</code>	34 MB	The full labelled modelling frame: 279,650 rows × 93 columns (subdivision × init × lead, with the bust label, all 52 features, and the split tag). This is the single file <code>train_model.py</code> needs.
<code>data/interim/imd_subdivisions.gpkg</code>	6 MB	The 34 modelled IMD subdivision polygons (built as unions of 641 Census-2011 districts). The API serves <code>/api/regions</code> from this.
<code>data/interim/truth_subdivision_daily.parquet</code>	180 KB	Observed subdivision-daily rainfall (area-mean), derived from the IMD gridded product.
<code>data/interim/weights_*.parquet</code>	~100 KB	Cached grid↔polygon area-overlap weights, keyed by grid geometry.
<code>data/artifacts/bust_model.joblib</code>	1.5 MB	The trained XGBoost booster + isotonic calibrator + feature list.
<code>data/artifacts/results.json</code>	7.5 KB	Held-out-year evaluation table (all baselines + model, per-lead, reliability).
<code>data/artifacts/*.csv</code>	<5 KB each	Ablation, SHAP importance, stress-test, and ENS-baseline comparison tables.
<code>data/artifacts/reference_distributions.npz</code>	134 KB	Training-year feature distributions, used by the drift monitor.

Derived-data provenance note: `dataset.parquet` and `truth_subdivision_daily.parquet` contain an observed-rainfall column derived from IMD's 0.25° gridded product (area-averaged to subdivisions, reformatted, and joined with model outputs). This is a transformative derivative work under IMD's research-use terms; attribution is in `LOGIC.md` §5. It is not a copy of the IMD product and cannot be used to reconstruct it.

2. What ships as a GitHub Release asset (~74 MB)

Asset	Size	What it is
<code>bulletins.sqlite</code>	74 MB	The precomputed bulletin store: 79,900 pre-scored (subdivision × init × lead) rows, each with the served bust probability (model + ENS spread since v0.2.0), the model alone, the ENS spread, bagged prediction interval, OOD status, SHAP reason strings, regime vector, and the baseline probability. This is what the offline dashboard serves.

It is a Release asset rather than a committed file because 74 MB of binary would bloat every clone of the git history. It is **fully regenerable** from the committed `dataset.parquet`, `bust_model.joblib` and `combiner.json` by running `scripts/generate_bulletins.py` (~100 s), so the Release is a convenience, not a dependency. Since v0.2.0 (D-030) the served probability combines the model with the 50-member ENS spread, so regenerating also needs the ENS spread for 2021 and 2022 on disk (`scripts/fetch_ens.py --year 2021` and `--year 2022` , about 7 MB of derived parquet under `data/raw/wb2/ens/` , gitignored).

Get it (either works):

```
# One command - downloads and verifies the checksum:
PYTHONPATH=src python scripts/fetch_release_artifacts.py

# ...or regenerate it locally from the committed artifacts:
PYTHONPATH=src python scripts/generate_bulletins.py
```

SHA-256 of `bulletins.sqlite` (v0.2.0): `deeac846b62e88e1169c349b75b4dacdd2a7f8b8b99191026de0bdc9f002a468`

Release v0.1.0 (the model-alone store, SHA-256 `5cbb61350c2ba63dbc2dc00bc1d13b9a3d2f9a70e66d8317121cf996e6f241c`) stays published. The API still serves it without error, with no combination shown and the landing page's combination sentence hidden.

3. What is NOT in the repo and never will be (~570 MB of raw data)

This is external third-party data. We do not rehost it — we cite it and reproduce it from the primary source. Rehosting it would be a licensing grey area and would teach the wrong lesson about how research artifacts are shared.

Dataset	Local size	Primary source	Fetch with
IMD 0.25° gridded rainfall (2016–2022)	170 MB	<code>imd pune.gov.in/cmpg/Griddata/Rainfall_25_NetCDF.html</code>	<code>scripts/fetch_imd.py</code>
WeatherBench 2 IFS HRES forecasts	72 MB (of ~9 GB transferred)	<code>gs://weatherbench2/datasets/hres/...</code> (anonymous GCS)	<code>scripts/fetch_hres.py</code>
WeatherBench 2 ERA5 analysis	~450 MB	<code>gs://weatherbench2/datasets/era5/...</code>	<code>scripts/fetch_era5.py</code>
WeatherBench 2 IFS ENS (50-member)	360 KB (subsampled)	<code>gs://weatherbench2/datasets/ifs_ens/...</code>	<code>scripts/fetch_ens.py</code>
Census-2011 district boundaries	10 MB	<code>github.com/datameet/maps</code>	<code>scripts/fetch_boundaries.py</code>

Full reproduction from nothing:

```
python scripts/fetch_imd.py
python scripts/fetch_hres.py
python scripts/fetch_era5.py
python scripts/fetch_ens.py --year 2019 --every 6
python scripts/fetch_ens.py --year 2020 --every 6
python scripts/fetch_ens.py --year 2022 --every 3
python scripts/fetch_boundaries.py
# then rebuild everything:
python scripts/build_truth.py
python scripts/build_dataset.py
python scripts/train_model.py
python scripts/generate_bulletins.py
```

Total download time is dominated by the WeatherBench 2 HRES pull (~20 min) because WB2's zarr chunks span the globe (you download the world and slice to India locally — see [DECISIONS.md](#) D-002).

4. The three tiers, at a glance

└ committed in git (~42 MB) ─────────┐ our derived products: dataset.parquet, trained model, geometry, eval tables	clone the repo -> tests pass, model loads, pipeline runs
└ GitHub Release asset (~74 MB) ─────────┐ bulletins.sqlite (precomputed dashboard)	one command -> offline dashboard runs
└ never in the repo (~570 MB, external) ─────────┐ raw IMD / WeatherBench 2 / Census data	scripts/fetch_*.py -> reproduce from source

This is the standard pattern for a reproducible-research repo: **cite the primary source, ship the reproduction script, host only your own derivatives**. It is what makes the repo credible rather than merely convenient.